

8.3 Feature Description

8.3.1 Subsystems

BQ769x0 consists of three major subsystems: Measurement, Protection, and Control. These work together to ensure that the fundamental battery pack parameters—voltage, current and temperature—are accurately captured and easily available to a host controller, while ensuring a baseline or secondary level of hardware protection in the event that a host controller is unable or unavailable to manage certain fault conditions.

备注

The BQ769x0 is intended to serve as an analog front-end (AFE) as part of a chipset system solution: A companion microcontroller is required to oversee and control this AFE.

- The Measurement subsystem's core responsibility is to digitize the cell voltages, pack current (integrated into a passed charge calculation), external thermistor temperature, and internal die temperature. It also performs an automatic calculation of the total battery stack voltage, by simply adding up all measured cell voltages.
- The Protection subsystem provides a baseline or secondary level of hardware protections to better support a battery pack's FMEA requirements in the event of a loss of host control or simply if a host is unable to respond to a certain fault event in time. Integrated protections include pack-level faults such as OV, UV, OCD, SCD, detection of an external secondary protector fault, and internal logic "watchdog" -style device fault (XREADY). Protection events will trigger toggling of the ALERT pin, as well as automatic disabling of the DSG or CHG FET driver (depending on the fault). Recovery from a fault event must be handled by the host microcontroller.
- The Control subsystem implements a suite of useful pack features, including direct low-side NCH FET drivers, cell balancing drivers, the ALERT digital output, an external LDO and more.

The following sections describe each subsystem in greater detail, as well as explaining the various power states that are available.

8.3.1.1 Measurement Subsystem Overview

The monitoring subsystem ensures that all cell voltages, temperatures, and pack current may be easily measured by the host. All ADCs are trimmed by TI.

ADC and CC data are always returned as atomic values if both high and low registers are read in the same transaction (using address auto-increment).

8.3.1.1.1 Data Transfer to the Host Controller

The BQ769x0 has a fully digital interface: All information is transferred through I²C, simply by reading and/or writing to the appropriate register(s) storing the relevant data. Block reads and writes, buffered by an 8-bit CRC code per byte, ensure a fast and robust transmission of data.

8.3.1.1.2 14-Bit ADC

Each BQ769x0 device measures cell voltages and temperatures using a 14-bit ADC. This ADC measures all differential cell voltages, thermistors and/or die temperature with a nominal full-scale unsigned range of 0 – 6.275 V and LSB of 382 μ V.

To enable the ADC, the [ADC_EN] bit in the SYS_CTRL1 register must be set. This bit is set automatically whenever the device enters NORMAL mode. When enabled, the ADC ensures that the integrated OV and UV protections are functional.

For each contiguous set of five cells (VC1 to VC5, VC6 to VC10), when no cells in that particular set are being balanced, each cell is measured over a 50-ms decimation window and a complete update is available every 250 ms. In the BQ76930 and BQ76940, every set of five cells above the primary five cells is measured in parallel. The 50-ms decimation greatly assists with removing the aliasing effects present in a noisy motor environment.